

Foundation (Jalapeno)

- Q1.** Which inequality represents the region below the line $y = 2x - 3$?
- (A) $y < 2x - 3$
(B) $y > 2x - 3$
(C) $y \leq 2x - 3$
(D) $y \geq 2x - 3$
(E) $y = 2x - 3$
- Q2.** A spaceship is traveling through space with a velocity of x km/h. The spaceship's distance from Earth is given by $y = 5x - 20$. If the spaceship is at least 50 km away from Earth, which inequality represents this situation?
- (A) $y \geq 50$
(B) $y \leq 50$
(C) $5x - 20 \geq 50$
(D) $5x - 20 \leq 50$
(E) $x \geq 14$
- Q3.** The cost of launching a satellite into space is given by the inequality $y > 3x + 1000$, where x is the weight of the satellite in kg. Which of the following graphs represents this inequality?
- (A) Solid line with y -intercept at $(0, 1000)$
(B) Dashed line with y -intercept at $(0, 1000)$
(C) Solid line with y -intercept at $(0, 0)$
(D) Dashed line with y -intercept at $(0, 0)$
(E) Dashed line with y -intercept at $(0, 1000)$ and shaded region above the line
- Q4.** A scientist is conducting an experiment in a lab, and the temperature of the experiment is given by the inequality $y < 2x + 5$. Which of the following is a possible solution to this inequality?
- (A) $(0, 0)$
- (B) $(0, 10)$
(C) $(5, 15)$
(D) $(10, 20)$
(E) $(10, 25)$
- Q5.** The boundary line of the inequality $y \leq x - 2$ is
- (A) Solid
(B) Dashed
(C) Horizontal
(D) Vertical
(E) Diagonal
- Q6.** The inequality $y > 4x - 1$ represents the region
- (A) Below the line $y = 4x - 1$
(B) Above the line $y = 4x - 1$
(C) On the line $y = 4x - 1$
(D) Below and on the line $y = 4x - 1$
(E) Above and on the line $y = 4x - 1$
- Q7.** Which inequality represents the region between the lines $y = 2x + 1$ and $y = 3x - 2$?
- (A) $2x + 1 \leq y \leq 3x - 2$
(B) $2x + 1 \geq y \geq 3x - 2$
(C) $3x - 2 \leq y \leq 2x + 1$
(D) $3x - 2 \geq y \geq 2x + 1$
(E) $2x + 1 < y < 3x - 2$
- Q8.** The solution region of the inequality $y < 3x + 2$ is
- (A) A half-plane below the line $y = 3x + 2$
(B) A half-plane above the line $y = 3x + 2$
(C) A half-plane on the line $y = 3x + 2$
(D) The line $y = 3x + 2$
(E) The point $(0, 0)$

Open Ended Questions (FRQ)

- Q9.** Graph the inequality $y > 2x - 5$ and shade the solution region. Label the x and y axes.
- Q10.** The cost of launching a satellite into space is given by the inequality $y > 4x + 2000$, where x is the weight of the satellite in kg. Graph this inequality and shade the solution region. Label the x and y axes.

Chef's Tips

- Chef's Tips
- Make sure to read each question carefully and choose the correct answer.
- Use a pencil to graph the inequalities and shade the solution regions.